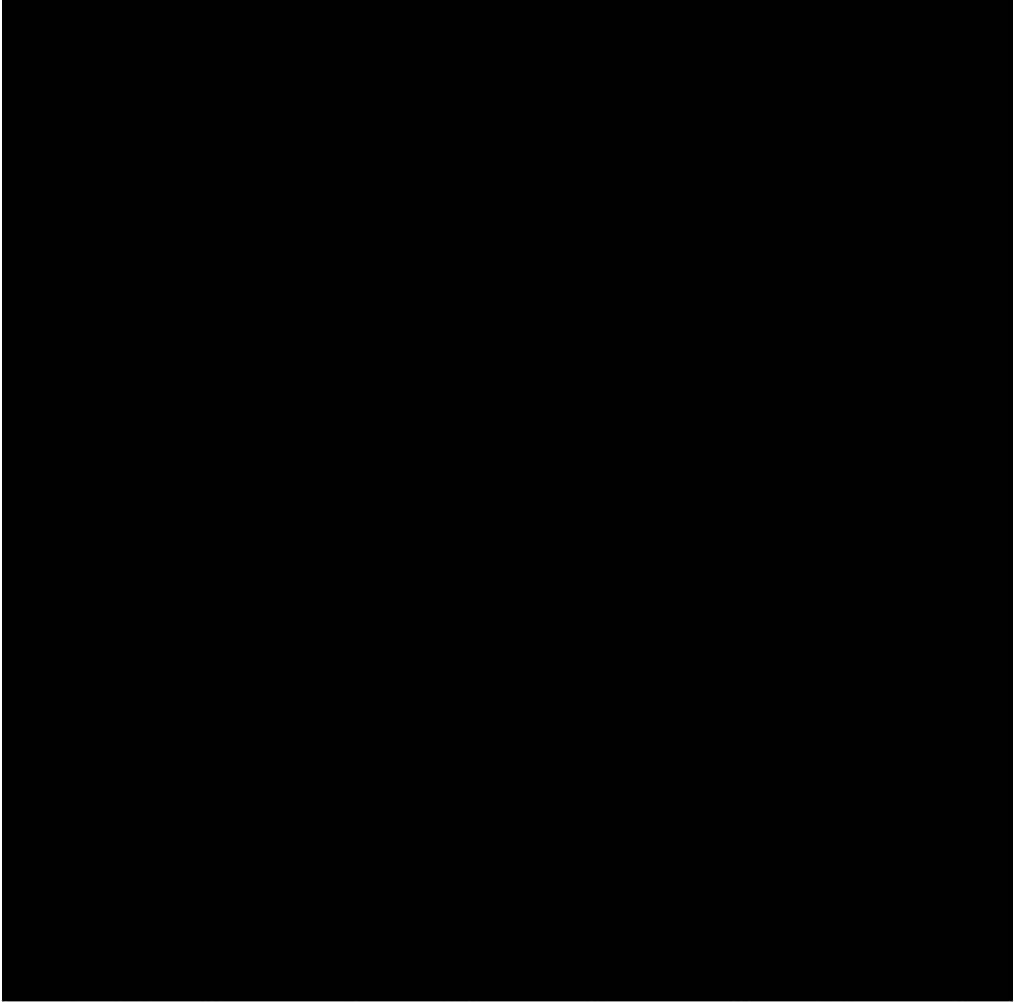
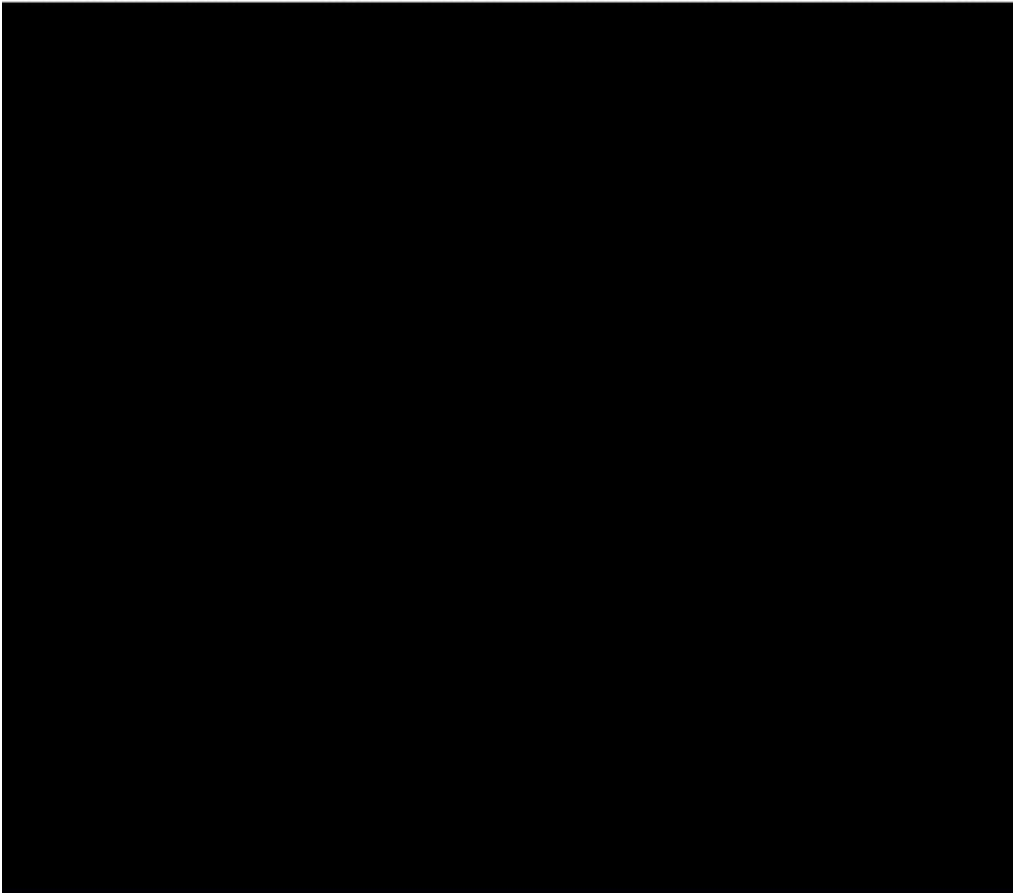
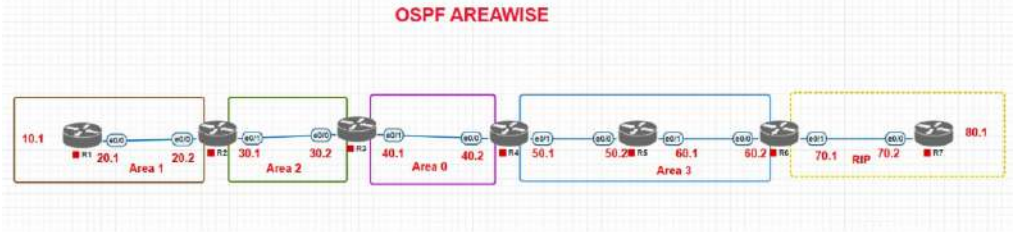




OSPF AREAWISE



```
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface ethernet0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#
Router(config)#interface ethernet0/0
Router(config-if)#ip address 20.0.0.0 255.0.0.0
Bad mask /8 for address 20.0.0.0
Router(config-if)#exit
Router(config)#interface ethernet0/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#
Router(config)#
*Oct  7 14:38:04.785: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct  7 14:38:05.785: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config)#
```

R1

```
*Oct 7 14:48:06.103: %SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 1
Router(config-router)#network 20.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#exit
Router#
*Oct 7 14:55:57.834: %SYS-5-CONFIG_I: Configured from console by console
Router#
*Oct 7 14:57:12.029: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/0 from
LOADING to FULL, Loading Done
Router#
```

```
Router(config)#
Router(config)#
Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 20.0.0.0 0.255.255.255 area 1
Router(config-router)#network 30.0.0.0 0.255.255.255 area 2
Router(config-router)#exit
Router(config)#
*Oct  7 15:00:33.889: %OSPF-6-AREACHG: 30.0.0.0/8 changed from area 1 to area 2
Router(config)#e
*Oct  7 15:00:35.181: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/1 from
LOADING to FULL, Loading Done
Router(config)#exit
Router#
*Oct  7 15:00:38.754: %SYS-5-CONFIG_I: Configured from console by console
Router#
```

R3

```
*Oct 7 14:52:47.350: %SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 30.0.0.0 0.255.255.255 area 2
Router(config-router)#network 40.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
*Oct 7 14:59:03.824: %SYS-5-CONFIG_I: Configured from console by console
Router#
*Oct 7 15:00:35.182: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/0 from
LOADING to FULL, Loading Done
Router#
*Oct 7 15:01:58.564: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Ethernet0/1 from
LOADING to FULL, Loading Done
Router#
```

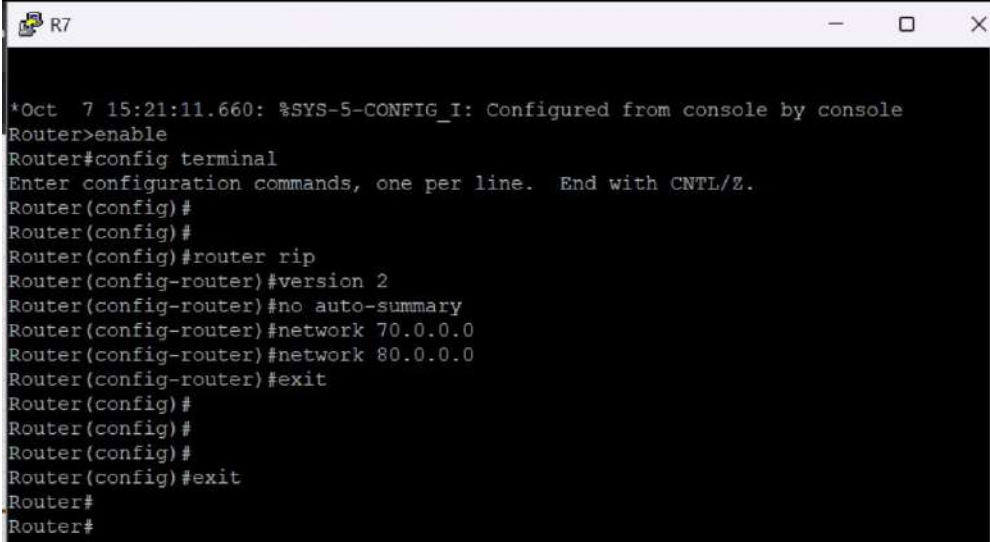
```
R4

*Oct  7 14:56:53.212: %SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#router-id 4.4.4.4
Router(config-router)#network 40.0.0.0 0.255.255.255 area 0
Router(config-router)#network 50.0.0.0 0.255.255.255 area
*Oct  7 15:01:58.564: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/0 from
LOADING to FULL, Loading Done
Router(config-router)#network 50.0.0.0 0.255.255.255 area 3
Router(config-router)#exit
Router(config)#exit
Router#
*Oct  7 15:02:06.780: %SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
*Oct  7 15:04:27.092: %OSPF-5-ADJCHG: Process 1, Nbr 5.5.5.5 on Ethernet0/1 from
LOADING to FULL, Loading Done
Router#
```

R5

```
*Oct  7 14:58:46.636: %SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#router ospf 1
Router(config-router)#router-id 5.5.5.5
Router(config-router)#network 50.0.0.0 0.255.255.255 area 3
Router(config-router)#netwo
*Oct  7 15:04:27.092: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Ethernet0/0 from
LOADING to FULL, Loading Done
Router(config-router)#network 60.0.0.0 0.255.255.255 area 3
Router(config-router)#exit
Router(config)#exit
Router#
*Oct  7 15:04:50.814: %SYS-5-CONFIG_I: Configured from console by console
Router#
*Oct  7 15:06:59.789: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on Ethernet0/1 from
LOADING to FULL, Loading Done
Router#
```

```
R6
*Oct  7 15:02:01.034: %SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#router-id 6.6.6.6
Router(config-router)#network 60.0.0.0 0.255.255.255 area 3
Router(config-router)#
*Oct  7 15:06:59.789: %OSPF-5-ADJCHG: Process 1, Nbr 5.5.5.5 on Ethernet0/0 from
LOADING to FULL, Loading Done
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#router rip
Router(config-router)#network 70.0.0.0
Router(config-router)#exit
Router(config)#
```



```
*Oct  7 15:21:11.660: %SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 70.0.0.0
Router(config-router)#network 80.0.0.0
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
Router#
_
```

```

R1
Router#
Router#
Router#
Router#show ip int br
Interface                IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0              20.0.0.1        YES manual up          up
Ethernet0/1              10.0.0.1        YES manual down        down
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1     FULL/BDR        00:00:36    20.0.0.2       Ethernet0/0

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.1/32 is directly connected, Ethernet0/0
Router#

```

R1

Router#

Router#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

Router#show ip ospf database

OSPF Router with ID (1.1.1.1) (Process ID 1)

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	1686	0x80000004	0x00BD37	1
2.2.2.2	2.2.2.2	1485	0x80000006	0x007B6E	1

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
20.0.0.1	1.1.1.1	1686	0x80000001	0x00F81C

Router#

OSPF AREAWISE

